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University of Toronto, Department of Physiology
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CURRENT POSITIONS/APPOINTMENTS

2019-present Mass Spectrometry Facility Manager/Senior Research Associate
Ted Rogers Centre for Heart Research, Toronto, ON, CA
University of Toronto, Department of Physiology, Toronto, ON, CA

2021-present Lecturer/Course Coordinator
University of Toronto, Department of Physiology, Toronto, ON, CA
PSL1040 - Systems Biology in Physiology covering topics in computational approaches to biomedical big data integration using statistics, bioinformatics, network biology and machine learning/AI

2023-present Board Member/Director
Access MS Scientific Inc., Markham, ON, CA
Mass Spectrometry and Bioinformatics Consulting

2024-present Board Member/Director
Pandecta Inc., Toronto, ON, CA
Data Analytics, Technology and Operations

2024-present NAIA Scientist
Institute for Clinical Evaluative Sciences (ICES), Toronto, ON, CA
Analysis of circulating biomarker profiles with ICES clinical data to develop predictive models for heart failure patient readmission using ML and AI analyses

EDUCATION

2008-2012 **Doctor of Philosophy**
Department of Laboratory Medicine and Pathobiology, University of Toronto
Supervisor: Dr. Eleftherios P. Diamandis
Topic: *Characterization of KLK6 glycosylation patterns in ovarian cancer; development of quantitative methods for detection of ovarian cancer specific KLK6 glycoforms; glycoproteomic characterization of ovarian cancer proximal fluids*

2003-2005 **Master of Science**
Department of Biochemistry, McGill University
Supervisor: Dr. Robert E. MacKenzie
Topic: *Characterization of the metabolic function of cytoplasmic methylenetetrahydrofolate Dehydrogenase-Cyclohydrolase-Synthetase activities*

1998-2002 **Honors Bachelor of Medical Science in Biochemistry (HBMSc)**
Department of Biochemistry, University of Western Ontario

PEER REVIEWED PUBLICATIONS (including submitted and in revision)

1. Martchenko A*, Lidington D*, Lin Z, Dinh DD, Radchenko AS, Kuzmanov U, Gramolini AO, Cox BJ, Bolz SS. ***Time-of-day and smooth muscle Bmal1-dependent multi-omic analysis of multiple microvascular beds reveals novel regulators of myogenic reactivity.*** Cell Reports. Revised manuscript (R1) under review; major revisions completed. *Equal contribution.
2. Webb BCW, Kuzmanov U, Gramolini AO, Wong E, Hofer SOP, Santerre JP. ***Electrospun multilayered gingival graft with a co-culture of adipose-derived cells in a gingival rat model.*** Biomaterials. Submitted.
3. Prajapati K, Baichoo S, Punwasi N, Sinha A, Huigen SHJ, Kuzmanov U, Diez Benavente E, Wang B, Maegdefessel L, Fish JE, Subasri V, Howe KL, et al. ***Epigenetic aging and plasma proteomic signatures of plaque vulnerability define a hidden high-risk population in asymptomatic carotid stenosis.*** Communications Medicine. Submitted (transferred from Nature Medicine).
4. Hamed O, Reitz CJ, Kuzmanov U, Gramolini S, Hu M, Halpern J, Billia F, Gramolini AO. ***Cardiac proteomic and phosphoproteomic profiling defines clinically relevant molecular subgroups in human heart failure.*** Nature Cardiovascular Research. September 25, 2026;5:1016–1033.
5. Webb BCW, Tran G, Devaraj K, Lindsay E, Kuzmanov U, Gramolini AO, Hofer SOP, Santerre JP. ***Engineering a multilayered micro-vascularized soft tissue graft: integrating polyurethane composite nanofibres with co-cultures of human adipose-derived endothelial and stem cells.*** Biomaterials. July 2026;330:124064.

6. Callaghan NI, Durland LJ, Chen W, Kuzmanov U, Miranda MZ, Ding Y, Mirzaei Z, Ireland RG, Reitz C, Gorman RA, Wang EY, Wagner K, Kim MM, Audet J, Santerre JP, Gramolini AO, Billia F, Radisic M, Mital S, Ellis J, Backx PH, Simmons CA. ***Advanced physiological maturation of human iPSC-derived cardiomyocytes using an algorithm-directed optimization of defined media components.*** Nature Communications. March 31, 2026;17:4625.

7. Di Paola M, Reitz CJ, Kuzmanov U, Jia K, Gramolini AO. ***Subcellular proteomic analyses reveal REEP5 knockdown in the mouse heart disrupts mitochondrial networks.*** Mol Cell Proteomics. February 9, 2026;25(3):101527.

8. MacQuarrie KD, D'Costa KA, Antonyshyn JA, Saini SK, Kuzmanov U, Bendeck MP, Gramolini AO, Hofer SOP, Santerre JP. ***Smooth muscle cell-like support cells accelerate the autologous endothelialization of a polyurethane scaffold for vascular engineering.*** Acta Biomaterialia. January 2026;210:398–412.

9. Teng ACT, Mehangrey D, Vandenbelt A, Vearncombe K, Callahan JD, Mistry P, Li W, Reitz CJ, Hamed O, Roche M, Kuzmanov U, Fish JE, Epelman S, Gramolini AO. ***Glyoxal is a superior fixative to formaldehyde in promoting antigenicity and structural integrity in murine cardiac tissues.*** J Mol Cell Cardiol Plus. May 11, 2025;12:100454.

10. Brahmbhatt DH, Scolari FL, Fung NL, Otsuki M, Lawler PR, Ross HJ, Kuzmanov U, Gramolini AO, Luk AC, Billia F. ***Heart failure decompensation with cardiogenic shock exhibits distinct sequential inflammatory profiles***. ESC Heart Fail. February 9, 2025;12(3):2077–2086.

11. Wu Q, Rafatian N, Wagner KT, Blamer J, Smith J, Okhovatian S, Aggarwal P, Wang EY, Banerjee A, Zhao Y, Nash TR, Lu RXZ, Portillo-Esquivel LE, Li CY, Kuzmanov U, Mandla S, Virlee E, Landau S, Lai BF, Gramolini AO, Liu C, Fleischer S, Veres T, Vunjak-Novakovic G, Zhang B, Mossman K, Broeckel U, Radisic M. ***SARS-CoV-2 pathogenesis in an angiotensin II–induced heart-on-a-chip disease model and extracellular vesicle screening.*** *PNAS*. July 5, 2024.

12. Hamidzada H, Pascual-Gil S, Wu Q, Kent GM, Massé S, Kantores C, Kuzmanov U, Gomez-Garcia MJ, Rafatian N, Gorman RA, Wauchop M, Chen W, Landau S, Subha T, Atkins MH, Zhao Y, Beroncal E, Fernandes I, Nanthakumar J, Vohra S, Wang EY, Valdman Sadikov T, Razani B, McGaha TL, Andreazza AC, Gramolini A, Backx PH, Nanthakumar K, Laflamme MA, Keller G, Radisic M, Epelman S. ***Primitive macrophages induce sarcomeric maturation and functional enhancement of developing human cardiac microtissues via efferocytic pathways.*** Nature Cardiovascular Research. May 7, 2024;3(5):567–593.

13. Reitz CJ, Kuzmanov U, Gramolini AO. ***Multi-omic analyses and network biology in cardiovascular disease.*** Proteomics. September 10, 2023.

14. Reitz CJ, Tavassoli M, Kim DH, Hadipour-Lakmehsari S, Shah S, Teng ACT, Emili A, Oudit GY[#], Kuzmanov U[#], Gramolini AO[#]. ***Integrative proteomics and phosphoproteomics of failing human left ventricle identifies etiology-specific phosphorylation of CTNNA3***. PNAS. May 9, 2023.

[#]co-corresponding author

15. Yerra CG, Batchu SN, Kaur H, Kabir G, Liu Y, Advani SL, Tran DT, Sadeghian S, Sedrak P, Billia F, Kuzmanov U, Gramolini AO, Qasrawi DO, Petrotchenko E, Borchers CH, Connelly KA, Advani A. *Pressure overload induces ISG15 to facilitate adverse ventricular remodeling and promote heart failure.* J Clin Invest. May 1, 2023.

16. Dufour CR, B'chir W, Perry MC, Xia H, Kuzmanov U, Gainullina A, Dejgaard K, Scholtes C, Ouellet C, Zuo D, Sanguin-Gendreau V, Guluzian C, Smith HW, Muller WJ, Audet-Walsh E, Sergushichev AA, Emili A, Giguère V. *Integrated multi-omics analysis of adverse cardiac remodeling and metabolic inflexibility upon ErbB2 and ERRa deficiency.* Communications Biology. Sep 12, 2022.

17. Kuzmanov U, Gawri R, Zelinka A, Russel KA, Gramolini AO, Kandel R. *Phosphoproteomic analysis of chondrocytes after short-term exposure to inorganic polyphosphate.* bioRxiv. July 31, 2021.

18. Wang EY, Kuzmanov U, Smith J, Dou W, Rafatian N, Lai BFN, Lu RXZ, Wu Q, Yazbeck J, Zhang XO, Sun Y, Gramolini AO, Radisic M. *An Organ-on-a-Chip Model for Pre-Clinical Drug Evaluation in Progressive Non-Genetic Cardiomyopathy.* J Mol Cell Cardiol. 2021 Jun 30;160:97-110.

19. Shah H, Hacker A, Langburt D, McFadden M, Zhang H, Kuzmanov U, Zhou YQ, Dewar M, Hussain B, Hinz B, Gramolini AO, and Heximer SP. *Myocardial infarction induces cardiac fibroblast transformation within injured and non-injured regions of the mouse heart.* J Proteome Res. 2021 May 7;20(5):2867-2881.

20. Lee SH, Hadipour-Lakmehsari S, Kim DH, Kuzmanov U, Shah S, Kislinger T, Sharma P, Oudit GY, Gramolini AO. *Bioinformatic analysis of membrane and membrane-associated proteins in murine cardiomyocytes and human myocardium.* Scientific Data. 2020 Dec 1;7(1):425.

21. Lee SH, Kim DH, Kuzmanov U, Gramolini AO. *Membrane proteomic profiling of the heart: Past, present and future.* American Journal of Physiology – Heart and Circulatory Physiology. 2021 Jan 1;320(1):H417-H423.

22. Kuzmanov U, Wang EY, Vanderlaan R, Kim DH, Lee SH, Hadipour-Lakmehsari S, Guo H, Zhao Y, McFadden M, Sharma P, Billia F, Radisic M, Gramolini A, Emili A. *Mapping signaling perturbations in myocardial fibrosis via the integrative phosphoproteomic profiling of tissue from diverse sources.* Nature Biomed Eng. 2020 Sep;4(9):889-900.

23. Hakem Zadeh F, Teng ACT, Kuzmanov U, Chambers PJ, Tupling AR, Gramolini AO. *AKAP6 and phospholamban colocalize and interact in HEK-293T cells and primary murine cardiomyocytes.* Physiol Rep. 2019 Jul;7(14): e14144.

24. Hu LZ, Goebels F, Tan JH, Wolf E, Kuzmanov U, Wan C, Phanse S, Xu C, Schertzberg M, Fraser AG, Bader GD, Emili A. *EPIC: software toolkit for elution profile-based inference of protein complexes.* Nature Methods. 2019 Aug;16(8):737-742.

25. Hadipour-Lakmehsari S, Driouchi A, Lee SH, Kuzmanov U, Callaghan NA, Heximer SP, Simmons CA, Yip CM, Gramolini AO. ***Nanoscale reorganization of sarcoplasmic reticulum calcium in pressure-overload cardiac hypertrophy visualized by dSTORM.*** Scientific Reports. 2019 May 27;9(1):7867.

26. Zhou J, Ma H, Wu Y, Lv X, Wang J, Liu S, Li D, Wang H, Yan Y, Luo N, Li Q, Xu H, Zhang Q, Yu L, Guo H, Kuzmanov U, Di L, Wu Q, Duan J. ***Lipidomic profiling of subchronic As_4S_4 exposure identifies inflammatory mediators as sensitive biomarkers in rats.*** Metallomics. 2019 Mar 20;11(3):576-585.

27. Ma H, Zhou J, Guo H, Shang E, Zhu Z, Kuzmanov U, Lv X, Di L, Yu B, Wu Q, Duan J. ***A strategy for the metabolomics-based screening of active constituents and quality consistency control for natural medicinal substance toad venom.*** Anal Chim Acta. 2018 Nov 15;1031:108-118.

28. Kuzmanov U, Guo H, Buchsbaum D, Cosme J, Isserlin R, Abbasi C, Sharma P, Gramolini AO, Emili A. *Global phosphoproteomic profiling reveals perturbed signaling in a mouse model of dilated cardiomyopathy*. PNAS. 2016 Nov 1;113(44):12592-12597.

29. Patel P, Kuzmanov U, Mital S. ***Avoiding false discovery in biomarker research.*** BMC Biochem. *2016 Jul 30;17(1):17.*

30. Guo H, Garcia-Vedrenne AE, Isserlin R, Lugowski A, Morada A, Sun A, Miao Y, Kuzmanov U, Wan C, Ma H, Foltz K, Emili A. ***Phosphoproteomic Network Analysis in the Sea Urchin Strongylocentrotus purpuratus Reveals New Candidates in Egg Activation.*** Proteomics. *2015 Dec;15(23-24):4080-95*.

31. Guo H, Isserlin R, Lugowski A, Kuzmanov U, Emili A. ***Large-scale label-free phosphoproteomics: from technology to data interpretation.*** Bioanalysis*. 2014 Sep 6(18):2403-20.*

32. Kuzmanov U, Emili A. ***Using phosphoproteomics to monitor disregulated signaling networks in cardiac disease preceding heart failure: where are we now?*** Bioanalysis. *2013 Dec 5(23):2863-6*.

33. Begcevic I, Kosanam H, Martínez-Morillo E, Dimitromanolakis A, Diamandis P, Kuzmanov U, Hazrati LN, Diamandis EP. ***Semiquantitative proteomic analysis of human hippocampal tissues from Alzheimer's disease and age-matched control brains***. Clin Proteomics*. 2013 May 1;10(1):5.*

34. Kuzmanov U, Emili A. ***Protein-protein interaction networks: probing disease mechanisms using model systems.*** Genome Medicine. *2013 Apr 30;5(4):37.*

35. Kuzmanov U, Kosanam H, Diamandis EP. ***The Sweet and Sour of Serological Tumour Biomarker Quantification.*** BMC Medicine*. 2013 Feb 7;11:31.*

36. Kuzmanov U, Musrap N, Kosanam H, Batruch I, Smith CR, Soosaipillai A, Diamandis EP. ***Glycoproteomic identification of sialylated proteins in ovarian cancer proximal fluids and cell lines.*** Clin Chem Lab Med. 2012 Dec 8:1-10.

37. Bayani J, Kuzmanov U, Saraon P, Fung WA, Soosaipillai A, Squire JA, Diamandis EP. ***Copy-Number and Expression Alterations of miRNAs in the Ovarian Cancer Cell Line OVCAR-3, and Its Impact on Kallikrein 6 Protein Expression.*** Clin Chem. *2013 Jan;59(1):296-305.*

38. Kuzmanov U, Smith CR, Soosaipillai A, Batruch I, Diamandis A, Diamandis EP. ***Separation of KLK6 glycoprotein subpopulations in biological fluids by anion-exchange chromatography coupled to ELISA.*** Proteomics. 2012 Mar;12(6):799-809.

39. Kuzmanov U, Jiang N, Smith CR, Soosaipillai A, Diamandis EP. ***Differential N-Glycosylation of kallikrein 6 derived from ovarian cancer cells or the central nervous system.*** Mol Cell Proteomics. 2009 Apr;8(4):791-8.

40. Christensen KE, Patel H, Kuzmanov U, Mejia NR, Mackenzie RE. ***Disruption of the mthfd1 gene reveals a monofunctional 10-formyltetrahydrofolate synthetase in Mammalian mitochondria.*** J Biol Chem. (2005) 280(9):7597-602.

SPEAKER

1. 2025. ***Deep proteomic profiling of paired myocardial and serum samples identifies candidate biomarkers of right ventricular failure in end-stage heart disease.*** Invited speaker, Seer Breakfast Symposium "Proteomic Breakthroughs in Cardiac Dysfunction and Aging-Related Decline", HUPO 2025, 24th World Congress of the Human Proteome Organization, Toronto, ON, Canada. November 10, 2025.

2. 2025. *(with Batzoglou S)* ***From Data to Discovery: Best Practices for Proteomic Data Analysis.*** Webinar, Labroots (sponsored by Seer Inc.). September 17, 2025.

3. 2020. ***Global phosphoproteomic profiling of signaling pathway aberrations in clinical samples, animal and organ-on-a-chip models.*** Keystone Symposium "Charting a New Course for Heart Failure: From Discovery to Data", Keystone, CO, USA.

4. 2018. *(with Hadipour-Lakmehsari, S)* ***Global phosphoproteomic profiling reveals perturbed cardiac signaling in dilated cardiomyopathy patients.*** HUPO 2018, 17th World Annual Congress of the Human Proteome Organization, Orlando, FL, USA.

5. 2017. ***Next Generation Cardiac Phosphoproteomics: from sample preparation to data interpretation in a mouse model of dilated cardiomyopathy.*** 9th International Symposium On Enabling Technologies (ETP 2017). Ottawa, ON, Canada.

6. 2016. ***Arendi Core Consulting and R&D overview***. 9th Shandong Overseas Talent Exchange and Cooperation Fair / Jinan Overseas High-Caliber Talent Fair and Project Conference. Dezhou/Jinan, Shandong Province, China.

7. 2016. ***Profiling of signaling cascade changes in heart pathology through phosphoproteomics.*** TRCHR Research Symposium, Toronto, ON, Canada.

8. 2011. ***Separation of KLK6 glycoprotein subpopulations in biological fluids by an ELISA-coupled anion exchange method.*** 4th International Symposium on Kallikreins and Kallikrein-related Peptidases, Rhodes, Greece

9. 2009. ***Differential N-glycosylation of Kallikrein 6 in Ovarian Cancer.*** Molecular Pathology Rounds, Samuel Lunenfeld Research Institute, Toronto, Canada

10. 2009. ***Differential N-glycosylation of kallikrein 6 in ovarian cancer ascites and cerebrospinal fluid.*** 100th Annual American Association of Cancer Researchers (AACR) Meeting, Denver, United States

11. 2009. ***Differential N-glycosylation of kallikrein 6 in ovarian cancer ascites and cerebrospinal fluid.*** 3rd International Symposium on Kallikreins and Kallikrein-related Peptidases, Munich, Germany

POSTER PRESENTATIONS AND ABSTRACTS

1. Hu M, Elbatarny M, Kuzmanov U, Eliathamby D, Reitz CJ, Manshaei R, Schwartz M, Simmons C, Chung JCY, Kim R, Punwasi N, Subasri V, Wang B, Jobling R, Gramolini AO, Ouzounian M. ***(2026) A Tissue-Derived Multiomic Signature of High-Risk Thoracic Aortic Disease Is Detectable in Plasma Years Before Acute Aortic Events***. American Association for Thoracic Surgery (AATS) Annual Meeting 2027 (submitted 2026).
2. Elbatarny M, Hu M, Reitz C, Kuzmanov U, Eliathamby D, Manshaei R, Schwartz M, Punwasi N, Chu V, Jobling R, Kim R, Chung J, Simmons C, Subasri V, Wang B, Gramolini A, Ouzounian M. ***(2026) Comprehensive Multiomic Analysis Reveals Novel Cytoskeletal Dysregulation Pathways in Thoracic Aortopathy Progression***. Canadian Cardiovascular Congress 2026. Ottawa, ON, Canada, October 15–17, 2026. Can J Cardiol 42(10 Suppl):S14 (P014).
3. Alibhai FJ, Sun G, Reitz CJ, Valdman Sadikov T, Kuzmanov U, Chung H, Ozcan B, Vatani P, Romero Torres D, Cortes-Medina LV, Qiang B, Amon C, Gramolini AO, Laflamme MA. ***(2026) An Electrical Stimulation Bioreactor for Scalable Human Pluripotent Stem Cell-Derived Cardiomyocyte Maturation***. ISSCR 2026, International Society for Stem Cell Research Annual Meeting.
4. Prajapati K, Baichoo S, Punwasi N, Kuzmanov U, Fish J, Subasri V, Howe K. ***(2026) Identification of High-Risk Asymptomatic Carotid Artery Stenosis Patients Using Multiomics and Molecular Aging Clocks***. AHA Vascular Discovery 2026: From Genes to Medicine Scientific Sessions. Bellevue, WA, USA, May 13–16, 2026. Arterioscler Thromb Vasc Biol 46(Suppl 1):Fri036.
5. Kuzmanov U, Davoudpour D, Gramolini A, Billia F. ***(2025) Deep Proteomic Profiling of Paired Myocardial and Serum Samples Identifies Biomarkers of Right Ventricular Failure in LVAD Patients***. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 11, 2025.
6. Lo KL, Reitz CJ, Hamed O, Davoudpour D, Kuzmanov U, Billia F, Gramolini AO. ***(2025) Phosphoproteome Dysregulation at the Cardiomyocyte Intercalated Disc in Human Heart Failure***. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 11, 2025 (oral presentation).
7. Di Paola M, Kuzmanov U, Subasri V, Abdul-Samad K, Wang B, Lee DS, Billia F, Gramolini A. ***(2025) Plasma Proteome Profiling in Acute Decompensated Heart Failure Reveals Biomarkers Predictive of Readmission Risk***. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 11, 2025.
8. Davoudpour D, Reitz CJ, Kuzmanov U, Jia K, Hamed O, Gramolini AO. ***(2025) Affinity-Purification Mass Spectrometry (AP-MS) and Kinase Inhibitor Screens Identify Phosphorylation-Mediated Regulation of αT-Catenin at the Cardiomyocyte Intercalated Disc***. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 10, 2025.
9. Di Paola M, Reitz CJ, Kuzmanov U, Jia K, Gramolini AO. ***(2025) Subcellular Proteomic Analyses of Reep5 Knockdown Reveal Disrupted Mitochondrial Dynamics in the Mouse Heart***. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 12, 2025.

10. Teng A, St. Germain J, Vandenbelt A, Nicholson S, Harding S, Kuzmanov U, Raught B, Gramolini A. *(2025) Proteomic Analysis of Protein Aggresomes Reveals ARMC6 as a Stress-Induced Marker of Extrachromosomal DNA in Cancer*. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 12, 2025.
11. Teng A, Mehangrey D, Vandenbelt A, Vearncombe K, Callahan JD, Mistry P, Li W, Reitz CJ, Hamed O, Roche M, Kuzmanov U, Fish JE, Epelman S, Gramolini AO. *(2025) Glyoxal Outperforms Formaldehyde in Preserving Protein Complexes for xIP-MS and Antigenicity for Immunostaining in Mouse Hearts*. HUPO 2025, 24th World Congress of the Human Proteome Organization. Toronto, ON, Canada, November 12, 2025.
12. Elbatarny M, Peterson S, Zubair A, Eliathamby D, Kuzmanov U, Chu V, Chung JCY, Wang B, Simmons C, Gramolini AO, Ouzounian M, on behalf of MultiTAAD. *(2025) Multiomic Tissue and Plasma Analysis Reveals Aortic Valve and Thoracic Aortopathy Phenotypes Are Synergistically Related*. Heart Valve Society Annual Meeting 2025.
13. MacQuarrie KD, D'Costa KA, Antonyshyn JA, Saini SK, Kuzmanov U, Gramolini AO, Hofer SOP, Santerre JP. *(2025) The Selection of Support Cells to Produce a Fully Endothelialized and Biomimetic Vascular Construct*. Toronto Biomedical Engineering Conference (ToBE) 2025.
14. Kuzmanov U, Hamidzada H, Wang E Y, Radisic M, Epelman S, Gramolini A. *(2024) Evolution of proteomics of bioengineered human cardiac microtissues*. World Annual Congress of the Human Proteome Organization. Dresden, Germany, October, 2024.
15. Kuzmanov U, Subasri V, Abdul-Samad K, Wang B, Lee D, Gramolini A, Billia P. *(2024) Comprehensive Plasma Proteomics in Heart Failure Patients: Findings and Future Directions.* World Annual Congress of the Human Proteome Organization. Dresden, Germany, October, 2024.
16. Teng A, St. Germain J, Kuzmanov U, Raught B, Gramolini A. *(2024) Developing a miniTurboID mass spectrometry-based method for characterizing aggregate proteomics in metabolically distressed cells*. World Annual Congress of the Human Proteome Organization. Dresden, Germany, October 23, 2024 (oral presentation).
17. Hamed O, Kuzmanov U, Gramolini S, Halpern J, Brahmbhatt D, Grothe D, Billia F, Gramolini A. *(2024) Proteomic profiling of human heart failure reveals specific and shared molecular responses across etiologies*. International Society for Heart Research North American Section (ISHR-NAS) Annual Meeting. Long Beach, CA, USA, August 2024. J Mol Cell Cardiol Plus 12(Suppl):100379.
18. Reitz C, Jia K, Alibhai F, Di Paola M, Au A, Kuzmanov U, Yip C, Laflamme M, Gramolini A. *(2024) Integral membrane-shaping proteins drive sarco(endo)plasmic reticulum formation and maturation during cardiomyocyte development*. International Society for Heart Research North American Section (ISHR-NAS) Annual Meeting. Long Beach, CA, USA, August 2024. J Mol Cell Cardiol Plus 12(Suppl):100377.
19. Davoudpour D, Reitz C, Kuzmanov U, Hamed O, Gramolini A. *(2024) Characterizing the mechanisms and implications of αT-catenin phospho-regulation at the cardiomyocyte intercalated disc*. International Society for Heart Research North American Section (ISHR-NAS) Annual Meeting. Long Beach, CA, USA, August 2024. J Mol Cell Cardiol Plus 12(Suppl):100417.
20. Di Paola M, Reitz CJ, Kuzmanov U, Jia K, Gramolini AO. *(2024) Sarco(endo)plasmic reticulum membrane protein REEP5 regulates mitochondrial dynamics in the mouse heart*. International Society

for Heart Research North American Section (ISHR-NAS) Annual Meeting. Long Beach, CA, USA, August 2024. J Mol Cell Cardiol Plus 12(Suppl):100427.

21. Elbatarny M, Kuzmanov U, Eliathamby D, Chu V, Nedadur R, Reitz CJ, Hamed O, Simmons CA, Chung JCY, Wang B, Ouzounian M, Gramolini AO. ***(2023) Distinct Proteomic Profiles in Human Thoracic Aortic Aneurysm by Segment May Drive Prognostic Differences: Conventional & Machine Learning Analysis.*** AHA Scientific Sessions. Philadelphia, USA. November 11–13, 2023.
22. Reitz CJ, Tavassoli M, Kim DH, Shah S, Lakin R, Teng ACT, Zhou Y, Li W, Hadipour-Lakmehsari S, Backx PH, Emili A, Oudit GY, Kuzmanov U, Gramolini AO. ***(2023) Phosphorylation Signatures of Human Heart Failure: Characterizing Dilated Cardiomyopathy-associated Signaling Pathways at the Cardiomyocyte Intercalated Disc.*** Basic Cardiovascular Sciences Scientific Sessions 2023. Boston, USA. July 31-Aug 3, 2023.
23. Di Paola M, Reitz CJ, Kuzmanov U, Jia K, Teng ACT, Gramolini AO. ***(2023) Confocal and Mass Spectrometry-based Investigation of REEP5 Depletion by AAV9 in the Mouse Heart.*** Basic Cardiovascular Sciences Scientific Sessions 2023. Boston, USA. July 31-Aug 3, 2023.
24. Davoudpour D, Reitz CJ, Kuzmanov U, Gramolini AO. ***(2023) Investigating αT-Catenin Phosphorylation and Evaluating its Cardioprotective Effects in Heart Failure.*** Frontiers in Physiology 2023, 43rd Annual Research Symposium, Department of Physiology, University of Toronto. May 26, 2023.
25. Di Paola M, Reitz CJ, Kuzmanov U, Jia K, Teng ACT, Gramolini AO. ***(2023) Confocal and Mass Spectrometry-based Investigation of REEP5 Depletion by AAV9 in the Mouse Heart***. Frontiers in Physiology 2023, 43rd Annual Research Symposium, Department of Physiology, University of Toronto. May 26, 2023.
26. Kuzmanov U, Reitz CJ, Wang EY, Tavassoli M, Kim DH, Hadipour-Lakmehsari S, Shah S, Teng ACT, Billia F, Emili A, Radisic M, Oudit GY, Gramolini AO. (2022) ***Global phosphoproteomic profiling of signaling pathway aberrations in human and mouse cardiomyopathies with differing etiologies***. HUPO 2022, World Annual Congress of the Human Proteome Organization, Cancun, Mexico.
27. Reitz CJ, Tavassoli M, Kim DH, Hadipour-Lakmehsari S, Shah S, Teng ACT, Emili A, Oudit GY, Kuzmanov U, Gramolini AO. (2022) ***Integrative proteomic and phosphoproteomic analysis identifies etiology-specific phosphorylation patterns in the failing human heart.*** XLI Annual Meeting of the International Society for Heart Research (ISHR)-North American Section. Winnipeg, Manitoba, Canada.
28. Di Paola M, Kuzmanov U, Reitz CJ, Teng ACT, Gramolini AO. (2022) ***Sarco(endo)plasmic reticulum membrane protein REEP5 regulates subcellular structure and function in the*** heart. XLI Annual Meeting of the International Society for Heart Research (ISHR)-North American Section. Winnipeg, Manitoba, Canada.
29. Kuzmanov U, Reitz CJ, Tavassoli M, Kim DH, Hadipour-Lakmehsari S, Shah S, Emili A, Oudit GY, Gramolini AO. (2022) ***Integrative proteomic and phosphoproteomic analysis of ischemic and dilated cardiomyopathy cardiac tissue from human explants.*** XXIV World Congress International Society for Heart Research (ISHR). Berlin, Germany.

30. Di Paola M, Kuzmanov U, Reitz CJ, Teng ACT, Gramolini AO. (2022) ***Sarco(endo)plasmic reticulum membrane protein REEP5 regulates subcellular structure and function in the heart.*** Frontiers in Physiology 2022, 42nd Annual Research Symposium, Department of Physiology, University of Toronto.

31. Reitz CJ, Tavassoli M, Kim DH, Hadipour-Lakmehsari S, Shah S, Lee SH, Teng ACT, Emili A, Oudit GY, Kuzmanov U, Gramolini AO. (2021) ***Identification of Novel Phosphoprotein Signaling Pathways in Human Dilated Cardiomyopathy by Integrative Proteomic and Phosphoproteomic Analysis.*** Basic Cardiovascular Sciences Scientific Sessions 2021 (virtual).

32. Di Paola M, Kuzmanov U, Teng ACT, Lee SH, Reitz CJ, Gramolini AO. (2021) ***Analysis of Receptor Expression-Enhancing Protein 5 (REEP5) on cardiac structural development and function using model systems.*** Frontiers in Physiology 2021, 41st Annual Research Symposium, Department of Physiology, University of Toronto.

33. Zhang H, Kuzmanov U, and the Human Explanted Heart Program (HELP) team. ***(2020) Distinct Remodeling Pattern Between Pediatric and Adult Heart Failure***. European Society of Cardiology (ESC) Congress 2020.

34. Kuzmanov U, Wang E, Vanderlaan R, Guo H, Isserlin R, Hadipour-Lakmehsari S, Sharma P, Billia F, Radisic M, Gramolini AO, Emili A. (2019). ***Integrative Phosphoproteomic Profiling of Clinical Samples, Animal and Organ-On-A-Chip Models: Mapping Fibrotic Signaling Cascades***. Keystone Symposium - Proteomics and its Application to Translational and Precision Medicine, Stockholm, Sweden.

35. Kuzmanov U, Vanderlaan, Guo H, Hadipour-Lakmehsari S, Sharma P, Billia P, Emili A, Gramolini AO. (2018). ***Global Human and Mouse Phosphoproteomic Profiling of Signaling Pathway Aberrations in Hypertrophic Cardiomyopathy.*** HUPO 2018, 17th World Annual Congress of the Human Proteome Organization, Orlando, FL, USA.

36. Kim DH, Kuzmanov U, Hadipour-Lakmehsari S, Emili A, Oudit G, Gramolini AO. (2018) ***Phosphopeptide Enrichment and Analysis of Human Ischemic Cardiomyopathic Tissues Reveal Infarct Versus Non-Infarct Unique signaling Pathways.*** HUPO 2018, 17th World Annual Congress of the Human Proteome Organization, Orlando, FL, USA.

37. Wang EY, Kuzmanov U, Zhao Y, Rafatian N, Gramolini AO, Emili A, Backx P, Radisic M. (2018). ***A Combined Strategy of Human Induced Pluripotent Stem Cells, Organ-on-a-chip Engineering and Precision Proteomics Analysis for Cardiac Disease Study.*** *5th TERMIS World Congress 2018, Kyoto, Japan.*

38. Kuzmanov U, Vanderlaan R, Hadipur-Lakmehsari S, Guo H, Buchsbaum D, Cosme J, Abbasi C, Isserlin R, Sharma P, Billia F, Emili A, Gramolini AO. (2017). ***Phosphoproteomic profiling of heart tissue: from mouse model to patient samples.*** 16th Human Proteome Organisation World Congress, Dublin, Ireland

39. Kuzmanov U, Guo H, Buchsbaum D, Cosme J, Abbasi C, Isserlin R, Sharma P, Gramolini AO, Emili A. (2017). ***Phosphoproteomics-based elucidation of disturbed signaling pathways in a mouse model of dilated cardiomyopathy.*** Heart Failure Update 2017, Toronto, Canada

40. Kuzmanov U, Guo H, Buchsbaum D, Cosme J, Abbasi C, Isserlin R, Sharma P, Gramolini AO, Emili A. (2017). ***Cardiac phosphoproteomics: profiling of a mouse model of dilated cardiomyopathy.*** Canadian National Proteomics Network Annual Meeting, Toronto, Canada

41. Kuzmanov U, Musrap N, Kosanam H, Smith CR, Batruch I, Diamandis EP. (2012). ***Glycoproteomic identification and characterization of potential glycoprotein biomarkers in ovarian cancer proximal fluids.*** American Society for Mass Spectrometry (ASMS) Annual Conference, Vancouver, Canada

42. Kuzmanov U, Musrap N, Kosanam H, Smith CR, Diamandis EP. (2011). ***Proteomic identification of sialic acid containing glycoproteins in ovarian cancer proximal fluids and cell line supernatants.*** Canadian Cancer Research Conference, Toronto, Canada

43. Bayani J, Kuzmanov U, Smith CR, Batruch I, Squire JA, Diamandis EP. (2011). ***Mechanisms of KLK6 Expression and Biomarker Discovery in Ovarian Carcinomas: An Integrated Genomic, microRNA (miRNA) and Proteomic Profiling Approach.*** Canadian Cancer Research Conference, Toronto, Canada

44. Kuzmanov U, Smith CR, Soosaipillai A, Batruch I, Diamandis A, Diamandis EP. (2011). ***ELISA coupled anion exchange method for separation of KLK6 glycoprotein subpopulations in biological fluids.*** 102nd Annual American Association for Cancer Researchers Meeting, Orlando, United States

45. Bayani J, Kuzmanov U, Smith CR, Batruch I, Presvelos J, Graham C, Squire JA, Diamandis EP. (2011). ***Integrated genomic, microRNA (miRNA) and proteomic profiling by stable isotope labeling with amino acids in cell culture (SILAC) of ovarian carcinoma for biomarker discovery.*** 102nd Annual American Association for Cancer Researchers Meeting, Orlando, United States

46. Kuzmanov U, Smith CR, Soosaipillai A, Diamandis EP. (2010*). N-glycosylation patterns of kallikrein 6 in ovarian cancer.* 5th Canadian Conference on Ovarian Cancer Research, Toronto, Canada

47. Bayani J, Kuzmanov U, Batruch I, Cho CK, Smith CR, Marrano P, Graham C, Butzow R, Katsaros D, Lin L, Zheng Y, Squire JA, Diamandis EP. (2010). ***Integrated Genomic MicroRNA (miRNA) and Proteomic Profiling of Ovarian Carcinoma for Biomarker Discovery.*** 5th Canadian Conference on Ovarian Cancer Research, Toronto, Canada

48. Bayani J, Kuzmanov U, Batruch I, Cho CK, Smith CR, Marrano P, Graham C, Butzow R, Katsaros D, Lin L, Zheng Y, Squire JA, Diamandis EP. (2010). ***Integrated Genomic, MicroRNA (miRNA) and Proteomic Profiling of Ovarian Carcinoma for Biomarker Discovery.*** 101th Annual American Association for Cancer Research Meeting, Washington, United States

49. Kuzmanov U, Jiang N, Smith CR, Soosaipillai A, Diamandis EP. (2008). ***Kallikrein 6 N-glycosylation in ovarian cancer ascites and cerebrospinal fluids.*** ASBMB meeting, Post-Translational Modifications: Detection and Physiological Evaluation, Lake Tahoe, United States

50. Kuzmanov U, Jiang N, Soosaipillai A, Diamandis EP. (2007). ***Differential glycosylation of ovarian cancer derived kallikrein 6.*** 2nd International Symposium on Kallikreins and Kallikrein-related peptidases, Santorini Island, Greece

51. Kuzmanov U, Jiang N, Soosaipillai A, Diamandis EP. (2007). ***Differential glycosylation of ovarian cancer derived kallikrein 6 and its potential use as a biomarker.*** 5th General Meeting of the International Proteolysis Society. Patras, Greece

AWARDS AND FUNDING

2026-present	Principal Investigator
	TBEP Seed Grant ($80 000)
	A Mass Spectrometry-based Modular Isobaric Peptide Tag Calibrator Platform for Absolute and Pseudo-absolute Protein Quantification in Plasma and Single Cell Cardiac Samples
	Source: Ted Rogers Centre for Heart Research
2024-present	Co-investigator
	Seer Bio Insights Grant ($100 000 USD)
	Cardiac tissue and plasma proteomics to identify biomarkers predicting right ventricular failure
	Source: Seer Bio Industry Partnership
2023-present	Co-investigator
	Precision Medicine Initiative ($500 000)
	Predicting Disease Trajectory in Patients with Heart Failure and Aortopathy
	Analysis of ~2000 Patient Blood Samples
	Source: Peter Munk Cardiac Centre, University Health Network
2020-2021	Principal co-applicant
	Collaborative Research Program for Drug Discovery ($200 000)
	Elucidation of Disrupted Signaling Pathways in HFpEF by Global Phosphoproteomic
	Analysis of Patients Tissue Explants
	Source: Akros Pharmaceutical
	Awarded but rejected by co-applicants
2015-2017	TRCHR Education Fund Fellowship ($80 000)
2011-2012	Laboratory Medicine and Pathobiology travel award ($500 per annum)
2011	Dr. Rajalakshmi S. Dittakavi and Dr. Prema M. Rao Graduate Award in Laboratory Medicine and Pathobiology ($1000)
2008-2011	University of Toronto Open Fellowship ($5000 per annum)
2008-2009	Dutkevich Foundation Travel Award ($500 per annum)
2000	The Richard and Julia Butler Scholarship, UWO ($500)
1998-2001	Faculty of Science Dean's Honor List, UWO
1998	The Western Scholar Award, UWO ($500)

MENTORING/SUPERVISION

Supervision of technical staff, undergraduate and graduate students including mentorship in aspects regarding design and implementation of research projects, data analysis, and preparation of figures, oral presentations and scientific manuscripts.

2022-present	Daniel Davidpour, MSc student
2021-2022	Christopher Oldfield, *PhD student*
2021-present	Malak Elbatarny, *MD-PhD student*
2019-2020	Parshant Loungani, *undergraduate student*
2018-present	Michelle DiPaola, *PhD student*
2017-2020	Frank Shin-Haw Lee, *PhD student*
2017-2019	Julia Da Hye Kim, *MSc student*
2017-2018	Sina Hadipour-Lakmehsari, *MSc student*
2015-2016	Diana Buchsbaum, *MSc student*
2012-2013	Tirth Patel, *4th year student*
2011-2012	William Fung, *4th year student*
2010	John Presvelos, *undergraduate summer student*
2009	Alissa Sperou, *undergraduate summer student*
2007-2009	Nianxin Jiang, *undergraduate summer and 4th year stude*nt

TEACHING EXPRIENCE

2021-present	University of Toronto, Department of Physiology, Course coordinator/Lecturer *PSL1040H Systems Biology in Physiology.*
2021	University of Toronto, Department of Anatomy, Lecturer *Lectured on mass spectrometry-based proteomics in biomedical research for ANA498.*
2019-present	University of Toronto, Faculty of Medicine, Lecturer *Lectured on network biology analysis in cardiovascular research for JCV3065.*
2018-present	University of Toronto, Department of Laboratory Medicine and Pathobiology, Lecturer *Held lectures on proteomics applications to study of post-translational modifications for LMP1535 and LMP1207.*
2005-2007	McMaster University, Department of Biochemistry, *Teaching Assistant* *Instructed a basic biochemistry lab course (3L03) to third year students for two semesters.*
2005	McGill University, Department of Biology, *Teaching Assistant* *Prepared and conducted laboratory exercises for a laboratory course in Cell and Molecular biology (BIOL 112).*
2004	McGill University, Department of Biology, *Teaching Assistant* *Conducted tutorial sessions for an introductory Molecular Biology course (BIOL 200).*

EMPLOYMENT HISTORY

2021-2022	Proteomics Lead and Consultant *Dalriada Therapeutics Inc., Mississauga, ON, CA* *Consulting in the design, execution, and analysis of proteomics experiments.*
2020-2023	Founder *LaraAna – Kuzmanov Consulting Inc., Mississauga, ON, CA* *Providing proteomics and bioinformatics consulting services for several drug development companies (Quthero Inc., Dalriada Therapeutics, Dunad Therapeutics, Janpix Bio)*
2016-2019	Co-founder/Chief Scientific Officer *Arendi Core Inc., Toronto, ON, CA* *Formerly an H2i and MaRs Ventures bio-incubator resident company* *Accepted into and completed the 2018-2019 "Entrepreneurship for Cardiovascular Health Opportunities" training program and funding competition* *2018 SVIEF-STAR pitch competition finalist*
2015-2018	Post-Doctoral Research Fellow *Ted Rogers Centre for Heart Research, Toronto, ON, CA* Supervisor: Dr. Anthony O. Gramolini Topic: Phosphoproteomic *and PTM study of cardiac pathologies in mouse models of disease and clinical samples.*
2012-2018	Post-Doctoral Research Fellow *Donnelly Centre for Cellular and Biomolecular Research, Toronto, ON, CA* Supervisor: Dr. Andrew Emili Topic: Mass *spectrometry-based and computational investigation of the roles post-translational modifications play in molecular signaling mechanisms.*
2007-2008	University of Toronto/Mt. Sinai Hospital, ***Research Assistant***
2005-2007	McMaster University, ***Research and Teaching Assistant***
2005	McGill University, ***Teaching Assistant***
2004	McGill University, ***Teaching Assistant***
1997–2007	Kuzmanov & Associates Inc., ***System Administrator***
1996	Oasis Technology Ltd., ***Assistant system administrator***

OTHER ACTIVITIES AND SKILLS

Language skills: Fluent (oral and written) in English and Serbo-Croatian, basic proficiency in Italian.

Professional Memberships: American Association for Cancer research, American Society of Mass Spectrometry, Human Proteome Organization, Canadian National Proteomics Network.

Journal Reviewer*:* International Journal of Cancer***,*** Clinical Biochemistry, Journal of Translational Medicine, ACS Biomaterials, Molecular and Cellular Proteomics, Canadian Journal of Cardiology Open